

Site Suitability Analysis for Solar Power Plant Using GIS and Multi-Criteria Decision Analysis(MCDA) for Kalahandi District, Odisha.

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Abstract:

The Indian state of Odisha has immense potential to tap into solar energy as a reliable source of renewable energy production to fuel its energy demands. In Kalahandi district, which is dominated by agriculture, decentralised solar energy projects can provide energy security and an income source to farmers who are increasingly facing income uncertainties due to shifting climatic patterns that affect their crop yields. This study uses Multi-Criteria Evaluation on satellite remote sensing data of Kalahandi district to determine suitable sites for developing solar power projects. The factors that were taken into consideration include Global Horizontal Irradiance (GHI), slope, aspect, altitude, distance to roads, and distance to built-up areas, wherein fuzzy analysis was used to determine suitability. Constraints (where solar projects absolutely cannot be developed) were taken to be water bodies, trees, flooded vegetation, built-up areas, and the Karlapat Wildlife Sanctuary area, including a 10-km buffer zone around it, in accordance with the Eco-Sensitive Zone mandates of the Government of India.

Using the Analytical Hierarchy Process to assign weights to the factors, the Weighted Sum method was used in ArcGIS Pro to generate a suitability map classified into 5 zones: least, less, sufficiently, moderately, and highly suitable. The result of the map indicated that approx 4,448 km² of the district is suitable, with 1,195 km² (15.09% of the total area) emerging as highly suitable. GHI was found to be generally high across all tehsils, from which it can be inferred that conventional crystalline silicon photovoltaic panels can be used for projects in this region.

1. Introduction:

One of such countries that has emphasized the goal of raising its use of renewables in its agenda for the forthcoming years is India as a nation. During COP26, it was set that by 2030, it should attain 500 GW of non-fossil fuel capacity (PIB Headquarters, n.d.). One of our major accomplishments, even before achieving our goal of having 49% of our installed power generation capacity from non-fossil fuel resources (International Energy Agency, 2025), has come true. For instance, there are massive solar power projects built by our countries like Bhadla (2.25 GW), located in Rajasthan; Pavagada (2 GW), located in Karnataka; and Kurnool (1000 MW), located in Andhra Pradesh, among others (Soleos, 2025).

However, the issue of energy poverty still rages in Indian society. Though initiative schemes such as Saubhagya and the Pradhan Mantri Ujjwala Yojana (PMUY) have reached upto 77 million households (Gupta et al., 2020), and power outages have declined from 4.2% to 0.1% over a period of ten years between 2014 and 2024 (International Energy Agency, 2025), there still exists a huge inequality in terms of access and use of this much-needed resource by 65% of all households, which can be classified as ‘more and most energy poor’ (Gupta et al., 2020). This implies that there is an inadequate supply of electricity accessible and accessible for a sufficiently long period of time to perform all necessary household chores.

In such cases, having enough electricity to carry out other tasks, such as running irrigation pumps or powering lights for agricultural fields or electric fences, is next to impossible. This will affect the productivity and economic stability of households that rely on agriculture or small industries such as small-scale manufacturing units for their livelihoods. Meanwhile, if households with more stable and affordable connections to the grid, like those in urban areas, decide to use high-voltage appliances like refrigerators, they put significant stress on the grid (up to 10%) and take up large quantities of electricity - increasing existing inequalities (Gupta et al., 2020).

In order for such an energy crisis to be solved, there is the option of decentralizing the provision of energy. This means the management of the supply of energy will belong to the community that is most in need of energy. Reducing reliance on the mains power supply source for energy and using small power supplies from solar roofs would greatly eliminate the suffering and poverty related to energy in rural settings (Powell, 2024).

In an energy-poor state like Odisha, the potential to create such decentralized power systems is high: according to the Ministry of New and Renewable Energy (MNRE), Odisha has the potential to meet 2.33% or 34.7GW of India’s renewable energy demands (Singh et al., 2023). The state experiences upward of 300 uninterrupted sunny days in a year, with the average solar input (insolation) similar to that of Karnataka at 300 W per square meter and the peaks of input similar to Gujarat at 900 W per square meter, a state that has optimally utilized its solar energy potential (Singh et al., 2023). Currently, nearly 77% of the state’s capacity (Singh et al., 2023) is fuelled only by coal, which needs to change since the world is moving away from fossil fuel-based energy sources due to the rapidly worsening climate crisis.

44% of Odisha’s workforce is employed in agriculture, but this sector only provides around 24% of the state's economic value, which shows that productivity is low due to low rates of technological advancement (CGIAR, 2025b). Solar installations can significantly improve this situation, while also helping reduce gender inequalities and aid women farmers in increasing their income (Aarti Kelkar Khambete, 2025). Kalahandi, identified as an aspirational district by the NITI Aayog, fares poorly on several key development parameters, starting with fair access to education, health, and nutrition, employment, digital connectivity, and the like (Iype, 2019). The current research proposal will enable the investigation of the scope of this district in undertaking solar installations through site suitability analysis and thus assist in the achievement of several goals of the United Nations’ Sustainable Development Goals, to wit: 1. No Poverty, 5. Gender

Equality, 7. Affordable and Clean Energy, 8. Decent Work and Economic Growth, 9. Industry, Innovation, and Infrastructure, 10. Reduced Inequalities, and 11. Sustainable Cities and Communities.

2. Literature Review:

The Aly paper (Aly et al., 2017) analyzes large-scale solar potential in Tanzania, using GIS and a multi-criteria decision-making (MCDM) approach. The researchers have analysed the suitability using the analytical hierarchy process (AHP) so that different criteria like solar radiation, slope, land-use, water bodies, grid proximity, and protected areas are given individual weightages. In the result, they presented high-resolution suitability maps of 1 km * 1 km for concentrated solar power (CSP) and PV photovoltaic plants. They identify hotspot zones where solar development will be most effective. This paper can help me combine GIS and AHP to identify the best solar sites in Kalahandi. This study will use similar methods to evaluate several factors and generate maps for each one.

The Charabi paper (Charabi & Gastli, 2011) uses a fuzzy logic-based GIS approach to identify suitable land for PV farms in Oman's desert environment. The authors have used fuzzy quantifiers and ordered weighted averaging (OWA) methods in their study. As per their results, only 0.5% of the area emerged as falling in the highly suitable category, which means that concentrated PV technology needs to be used to get the maximum efficient output from the solar panels. If the entire suitable area is used with maximum efficiency, then Oman can satisfy 45.5 times its present total energy demand. This shows the power of using high-efficiency solar panels, which can capture large amounts of the solar radiation that falls on them, but these will be less feasible to install in a country like India, where rural communities need them the most but do not have high budgets.

This paper will help to understand that the fuzzy multi-criteria decision approach can help manage data uncertainty and minimize errors in final predictions. If there is any data limitation or unavailability for the Kalahandi district - an area where not much scientific research has been conducted, and historically, investments in any kind of development or infrastructure have been low - This study will use the fuzzy approach to generate an accurate analysis of suitability even with incomplete or inadequate data.

The Adeboboye paper (Adeboboye et al., 2024) uses GIS remote sensing, AHP, and weighted linear combination methods to analyse solar potential in the Enugu state of Nigeria. The study used solar radiation, slope, accessibility, and land use parameters. Based on the analysis, the result they found was that 13.39% of the area is highly suitable, 33.82% moderately suitable, and 52.79% least suitable. In an area where land availability is limited, having such categories can help determine areas where small-scale solar projects can still be developed to meet the demands of small regions, even if the land is not considered highly suitable. This might increase repair

costs or the risk of damage to panels or poor productivity of panels, but after all, it is essential to meet demand in one way or another, even in sub-optimal ways, especially in energy-poor regions. This paper has given me real-world examples from a developing nation where the terrain and socio-economic conditions are similar to rural districts in India. The methodologies chapter has aided, how to make use of land classification techniques to classify various areas of land in terms of the percentages of suitability and make a map for the region of interest.

The Halder paper (Halder et al., 2022) identifies twelve factors, like solar radiation, slope, soil, land use, road and grid proximity, etc., and uses weighted overlay analysis (WOA) and AHP to assess the solar suitability of land in Kolkata and nearby areas. It classified the lands into two suitability categories: high and very high. This paper is the most relevant for my research topic since the study has been conducted within India, and the weighting techniques will be useful to apply to my own analysis. The urban-rural mapping framework used in the paper will help me to identify suitable solar clusters within my region. The advantage of identifying such clusters is that we can have a larger number of panels spread over a contiguous area instead of having only a few panels in small areas, broken up by other kinds of land use. Applying extra criteria such as continuous areas of land - instead of only looking at the total amount of land - can also help communities share solar resources and pool in their money for installation, upkeep, and repairs. While it is alright to have even small suitable areas where only one or a few families at a time can get electricity, identifying slightly larger areas has significant benefits in the long run.

3. Research questions:

1. What is the total area available in the Kalahandi district that is suitable for solar projects?
2. What percentage of the total land of the district is highly suitable?
3. What are the variations in solar radiation received in different areas of the district, and how can we determine which type of PV panels to install to get the maximum productivity based on the particular features of each region (for example, regular, high concentration)?

4. Study Area:

The Kalahandi district lies in the south-west part of the State of India, Odisha; its geographic location measures approximately 7920 km². This district is located between 19.3°N and 21.5°N latitudes and 82.20°E and 83.47°E longitudes. The district is bordered by Nuapada and Balangir to the north, Rayagada to the southeast, Nabarangpur to the south, and Kandhmal to the east. The administrative headquarters of the district is Bhawanipatna, which is the district's main urban and economic hub.

The district has diverse topography in its eastern part, the hilly terrains of the Eastern are situated, whereas broad plains and valleys are located in its western and central parts. The elevation of the district ranges between 122m to 1498m above mean sea level (*Kālāhandi Topographic Map, Elevation, Terrain, n.d.*), and its higher altitude areas are largely covered by forest, whereas low-lying plains are mainly used for agriculture. In the southern part of the district, the Karlapat Wildlife Sanctuary is located. which constitutes the major protected area, and in this study, it is used as an environmental constraint with a 10km buffer zone.

Kalahandi experiences a tropical monsoon climate with hot summers and moderate winters. The districts get an average rainfall of about 1200mm- 1400 mm. However, the region also experiences good solar irradiance throughout the year, averaging between 5.2 to 5.6 kWh per square meter per day (*Odisha – IFOREST, n.d.*). This factor makes Kalahandi a promising site for utilizing solar energy, especially in its barren lands and open plains.

Kalahandi represents a rural agrarian landscape, where settlements is very dispersed and limited industrial activity. This spatial setup gives unique opportunities to set up the solar infrastructure in a way that increases energy access, and ecological disruption can be minimized. The flat terrain of the district, high solar potential, and available non-forestland make it the optimal case for solar projects.

This study, therefore, centers on the entire Kalahandi district boundary, excluding environmentally sensitive zones such as built-up areas, forests, water bodies, crop land, and

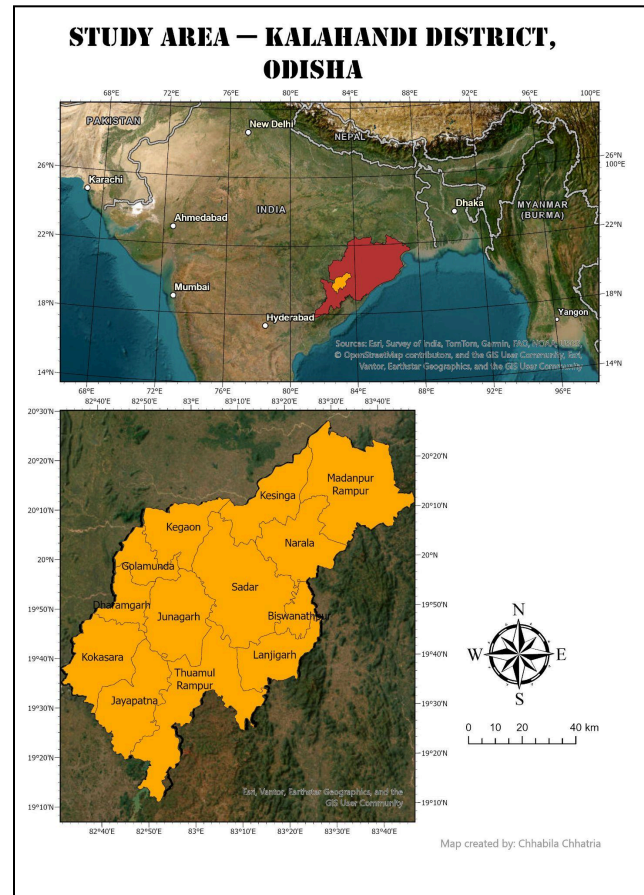


Figure 1. Study Area map

protected regions to assess and map the potential land for solar energy development using AHP-based GIS examination.

5. Materials

The study was conducted to select the best site for solar projects in Kalahandi's entire Tehsil area using the MCDA approach and AHP, a pairwise comparison matrix. Six parameters or criteria were applied to examine the best site for solar power projects in the Kalahandi district—ArcGIS Pro software with the weighted sum (spatial analyst). The tool was applied for the assessment(*How Weighted Sum works—ARCGIS Pro | Documentation*, n.d.). For the weighted sum approach, all the criteria were varied on a scale from low (1) to high(5) suitability classes, and AHP-based weighted values were used for the Weighted Sum analyst ArcGIS tool.

6. Data Used

The study used a combination of raster and vector datasets obtained from both Institutional archives and open source. The were chosen based on their spatial accuracy, resolution, and relevance for the site suitability assessment. To maintain spatial uniformity, all data setss were projected to the WGS 1984 UTM Zone 44N coordinates system to keep spacial uniformity. The data set details are provided in the table below.

Table 1. *Data used in the Study*

Criteria	Data source	Website
Land Use/ Land Cover(LULC)	Sentinel-2(ESRI Basemap)	https://livingatlas.arcgis.com/landcoverexplorer/#mapCenter=14.58892%2C51.86787%2C11.00&mode=step&timeExtent=2017%2C2024&year=2024
Global Horizontal Irradiance(GHI)	Global Solar Atlas	https://globalsolaratlas.info/map?c=19.837445.83.047714.11&s=19.850686.83.047714&m=site
Digital Elevation(DEM)	SRTM(USGS Earth Explorer)	https://earthexplorer.usgs.gov/
Slope, Aspect, Altitude	Derived From DEM In ArcGIS	https://earthexplorer.usgs.gov/
Distance From Roads	Open Street Map	https://www.geofabrik.de/data/download.html
Distance From Built-Up	Derived From LULC Layer	https://livingatlas.arcgis.com/landcoverexplorer/#mapCenter=14.58892%2C51.86787%2C11.00&mode=step&timeExtent=2017%2C2024&year=2024
Protected Area(Karlapat WLS)	World Database on Protected Areas (WDPA)	https://www.protectedplanet.net/en/thematic-areas/wdpa?tab=WDPA

Administrative Boundary(Kalahandi Dist)	Survey of India	https://www.surveyofindia.gov.in/
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To ensure the consistent pixel size during multi-criteria overlay operations, each raster layer was resampled to 30 m spatial resolution. All layers, such as Slope, Aspect, Altitude, GHI, Distance to Builtup, and Distance to Road, were normalized and reclassified based on suitability scores ranging from 1 (least suitability) to 5 (High Suitability). Water bodies, Forests, Flooded Vegetation, built-up areas, and Protected area Buffer, all as a constant layer, were allocated a value of 0 to exclude them from the final suitability map.

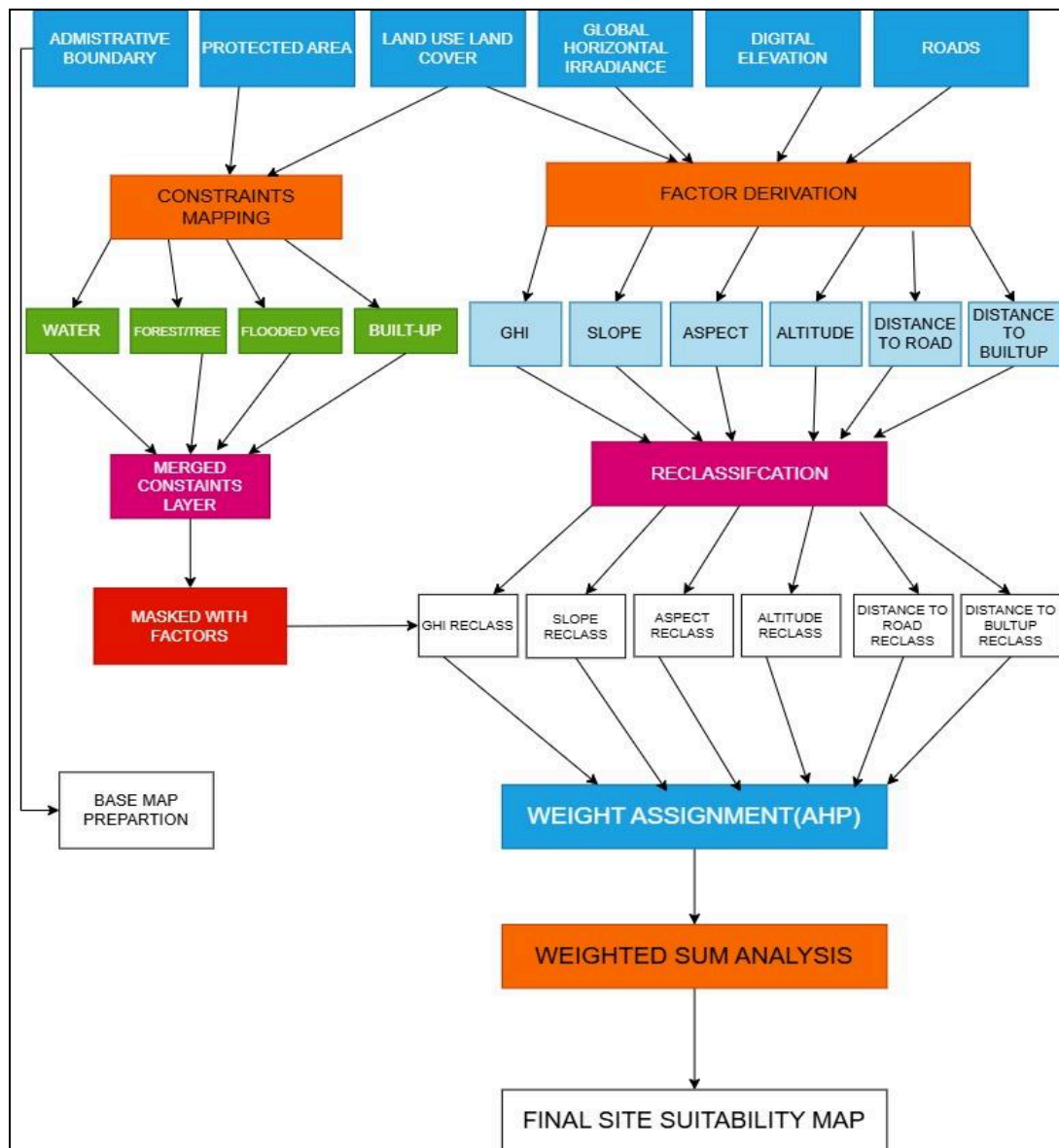
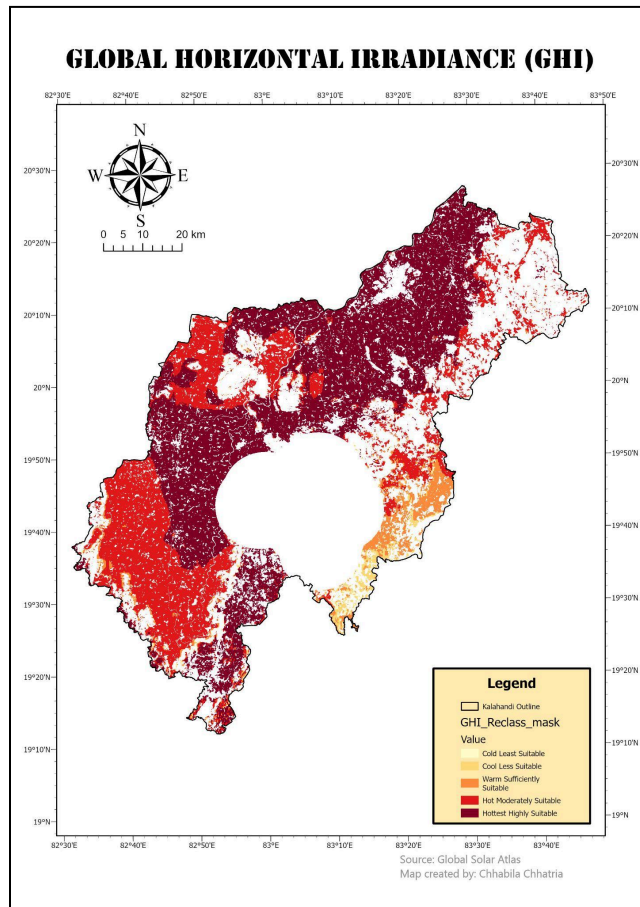


Figure 2. Detailed methodology of this study

7. Preparation and Computation Criteria

7.1) GHI- Global Solar Irradiance:

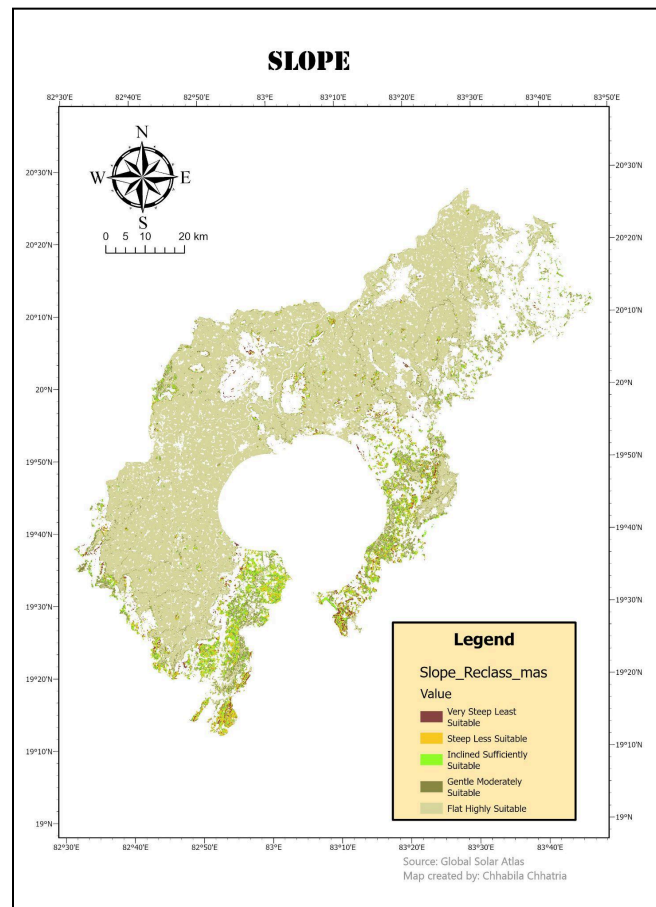


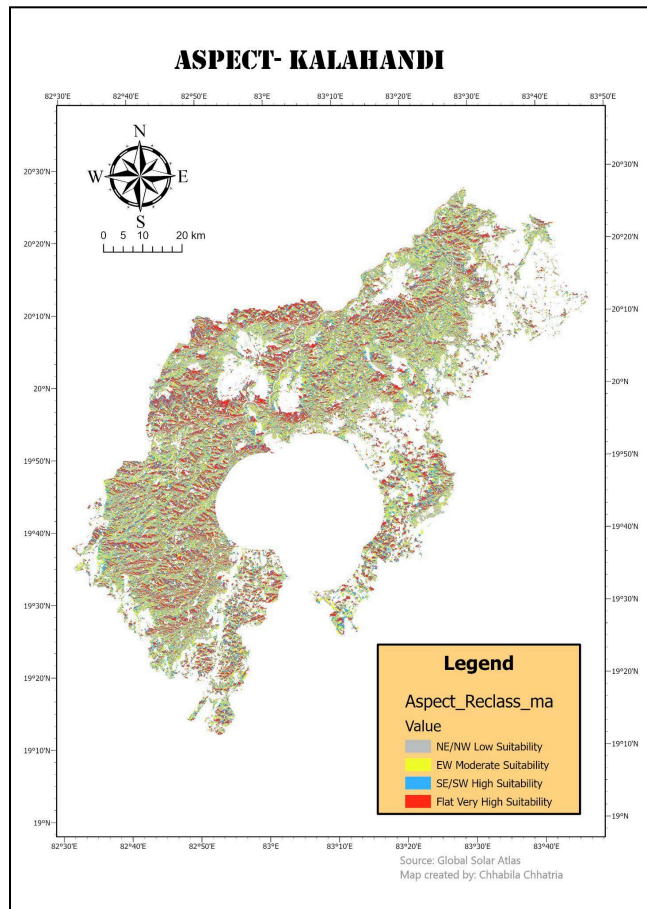
Solar radiation was the most affected and a crucial criterion for the area selection for the solar project. The data was taken from Global Solar Atlas. And the spatial resolution of the data is 1km. The raster was clipped to the study boundary(Kalahandi district) and standardized to kWh/m²/day. The areas that receive higher solar irradiance were examined as favorable for solar power generation. The solar radiation rate was separated into five classes: least suitable, less suitable, sufficiently suitable, moderately suitable, and highly suitable. This is one of the main factors for the analysis of site selection.

7.2) Slope:

The slope layer was derived from the DEM using the surface to slope tool. Flat/gentle slopes are best for the solar installation because they make construction easier.

The slope raster was reclassified into five classes: steep least, steep less suitable, inclined sufficiently suitable, gentle moderately suitable, flat highly suitable.



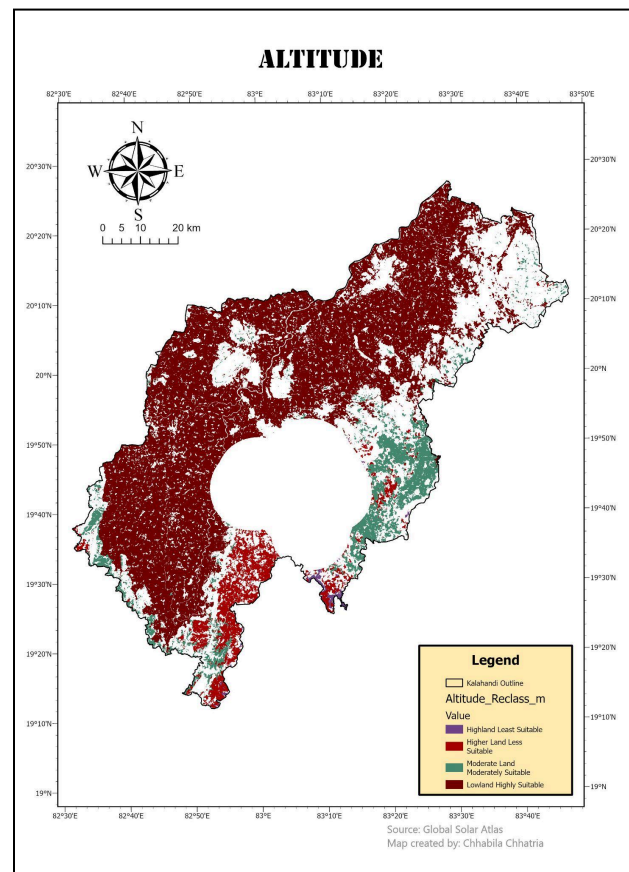


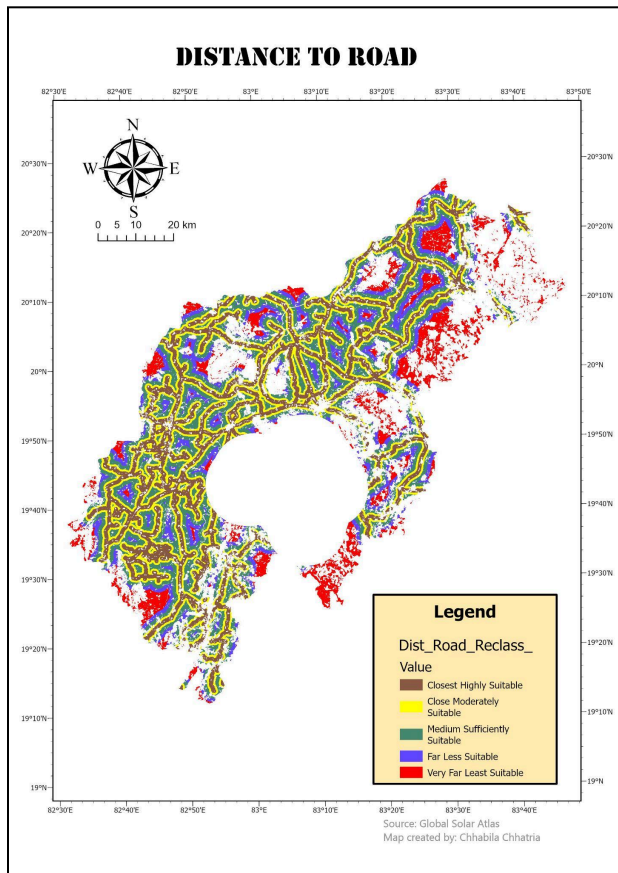
7.4) Altitude:

Altitude is extracted from the DEM. Altitude affects a solar project's feasibility indirectly. In very high elevation areas terrain is rugged, with low accessibility, and shading and environmental sensitivity are higher. In the Kalahandi district central plains elevation range is generally 200-400m, which provides suitable conditions for solar project installation. Based on the local conditions of Kalahandi, the suitability zones are reclassified into the highland-least suitable, higherland-less suitable, moderate land-moderately suitable, and lowland-highly suitable.

7.3) Aspect:

The aspect of the area was computed from the DEM to examine slope orientation. Due to the higher maximum solar exposure, also area was also assigned into four categories, such as Flat/South[135–225° & -1] very high suitability, SE/SW[90–135° & 225–270°] high suitability East/West[45–90° & 270–315°] moderate suitability, NE/NW[0–45° & 315–360°] low suitability.



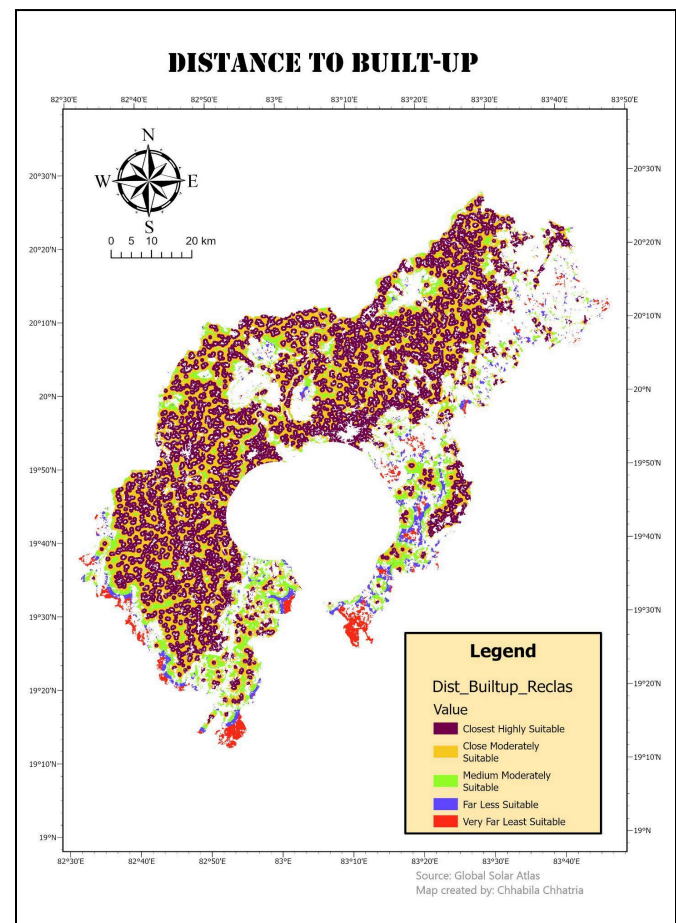


7.6) Distance to Built-up:

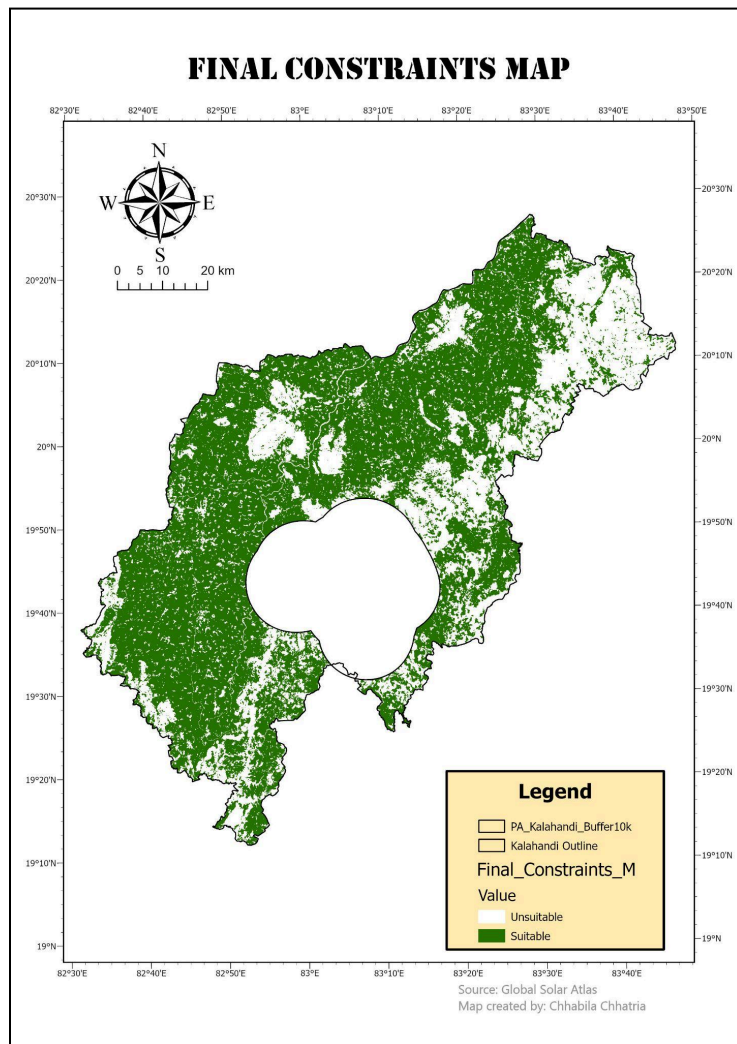
The spatial data of the transmission line was not available, so the built-up area to distance was considered as a proxy data as the electric infrastructure. It is assumed that the power distribution network is required to surround built-up areas, where it will be comparatively easier to connect the solar plant to the grid connection. Built-up areas were derived from Sentinel LULC data and converted into raster format, and then, using the Euclidean distance tool, the distance of each pixel from the nearest built-up area was calculated. And then

7.5) Distance to Road:

A road network is essential for any large-scale project, particularly for logistics and maintenance. This makes it easy to transport construction materials, cost-effective for maintenance, and feasible for grid connection. In this study, the distance from the road was calculated by using the tool: Euclidean Distance, because it provides the shortest physical distance. And then reclassified into five categories, such as closest- highly suitable, Closer-Moderately Suitable, Medium-Sufficiently Suitable, Far-Less Suitable, and Very Far Least Suitable.



reclassified into five classes, such as closet-high suitability, close-moderately suitable, Medium-Moderately Suitable, Far-less suitable, Very Far-Least Suitable.



7.7) Final Constraints Map:

For the solar power plants' site selection, there are certain land categories where development is not allowed due to social, environmental, or technical reasons.

Hence, this study followed a constraints mapping approach where unsuitable areas were excluded from the analysis from the outset.

- Certain land cover classes were considered permanent constraints. Using the Sentinel-2-based LULC data set. Because there are ecologically sensitive zones and land use conflicts, the solar power development is considered unsuitable.

a) LULC classes that are unsuitable are assigned as (0)

- Forest/ tree cover: because of biodiversity protection and legal restrictions.

- Flooded

Vegetation: because of instability and seasonal waterlogging.

- Water Bodies: Because of technical infeasibility and ecological importance.
- Built-up areas: because of land use conflict and existing settlements.

And other land cover classes, such as barren land and open agricultural land, are considered suitable and assigned a value of 1.

b) Protected Area constraints;

To ensure environmental protection, Karlapat Wildlife Sanctuary(*Protected Area | Wildlife Odisha*, n.d.) is considered a strict constraint. To prevent disturbance to

wildlife and the sensitive ecosystem zone, a 10km buffer zone has been created around the sanctuary. According to the government guidelines(Sanctuary Asia, 2021)

- c) Final Constraints Map preparation: The protected area buffer constraints and the land use and land cover constraints were merged using the Raster calculator, and the restricted area was assigned the value(0). Suitable areas were assigned a value(1), and outside the study boundary was considered as NoData.

Hence, the final constraints mask was applied to all reclassified factors rasters(Slope, GHI, Aspect, distance to road, altitude, and distance to built-up) using the raster calculator and SetNull function. It ensured that restricted areas' suitability scores would not be considered in the final analysis, and the Weighted Sum analysis could be performed only for Permissible Land parcels.

7.8) Weight Assignment using AHP(Analytic Hierarchy Process)

In the Multi-Criteria Evaluation(MCE) Framework, the importance of different factors is not the same. Therefore, in this study, if the weights are assigned intuitively or randomly, the result may be biased. Therefore, in this study, Analytic Hierarchy Process(AHP), which is a mathematical, structured, and consistency-based decision-making technique (What Is the Analytic Hierarchy Process (AHP)? | 1000minds, n.d.).

By using the AHP process, the relative importance of the criteria was carefully evaluated, which served as input for the weighted sum analysis.

The criteria were compared pairwise using a 1–9 scale, where:

Table 2: The importance scale value matrix

Important rank	Definition
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong importance
9	Extreme importance
2,4,6,8	Intermediate

During the comparisons, the solar project's feasibility, the physical conditions of the Kalahandi district, and some literature review were taken into consideration.

Table 3: Pairwise Comparison matrix.

Criteria	GHI	slope	Distance to road	Distance to built-up	Altitude	Aspect
GHI	1	3	5	5	7	7
Slope	1/3	1	3	3	5	5
Distance to Road	1/5	1/3	1	2	3	3
Distance to Bultup	1/5	1/3	1/2	1	3	3
Altitude	1/7	1/5	1/3	1/3	1	2
Aspect	1/7	1/5	1/3	1/3	1/2	1

The matrix was normalized by calculating the sum of each column of the pairwise matrix. After that, the average of each row was taken, which represents the principal eigen vector.

Table 4: Column Sum Calculation

Column	Sum
GHI	2.01
Slope	5.07
Distance to Road	10.17
Distance to Built-up	11.67
Altitude	19.50
Aspect	21.00

Table 5: Standardization of the matrix and Weights

Criteria	GHI	Slope	Distance to Road	Distance to Built-up	Altitude	Aspect	Row Average(Weight)
GHI	0.50	0.59	0.49	0.43	0.36	0.33	0.33
Slope	0.17	0.20	0.29	0.26	0.26	0.24	0.25
Road	0.10	0.07	0.10	0.17	0.15	0.14	0.15
Distance to Built-up	0.10	0.07	0.05	0.09	0.15	0.14	0.14
Altitude	0.07	0.04	0.03	0.03	0.05	0.10	0.08
Aspect	0.07	0.04	0.03	0.03	0.03	0.05	0.05

The Row average symbolizes the final AHP weight for each criterion.

Table 6: Final Weights Used in the Study

Criterion	Weight
GHI	0.33
Slope	0.25
Distance to Road	0.15
Distance to Built-up	0.14
Altitude	0.08
Aspect	0.005
Total	1.00

Consistency Check: Just a normal judgment of a human can be inconsistent and biased; therefore, the consistency ratio(CR) was calculated in AHP

- $CI = \frac{\lambda_{max} - n}{n - 1}$

- $CR = CI / RI$

Where CI indicates the consistency index, and λ_{max} denotes the maximum value of the pairwise comparison matrix approach

- n= Six criteria
- RI= Random Index(from Saaty Table)

Hence, the $CR < 0.10$, which shows that the pairwise comparison is consistent and reliable.

Integration of AHP with MCE:

The weights derived from the Analytic Hierarchy Process were used in the MCE Analysis. Each reclassified and its corresponding weight, and all layers were combined using the weighted sum tool.

7.9) The Weighted Sum Analysis:

Once the final weights for all criteria were derived through AHP, the weighted sum method was used to integrate these weights with the spatial layers. The weighted sum approach was chosen in this study because it is a transparent, widely accepted, and simple technique for MCE.

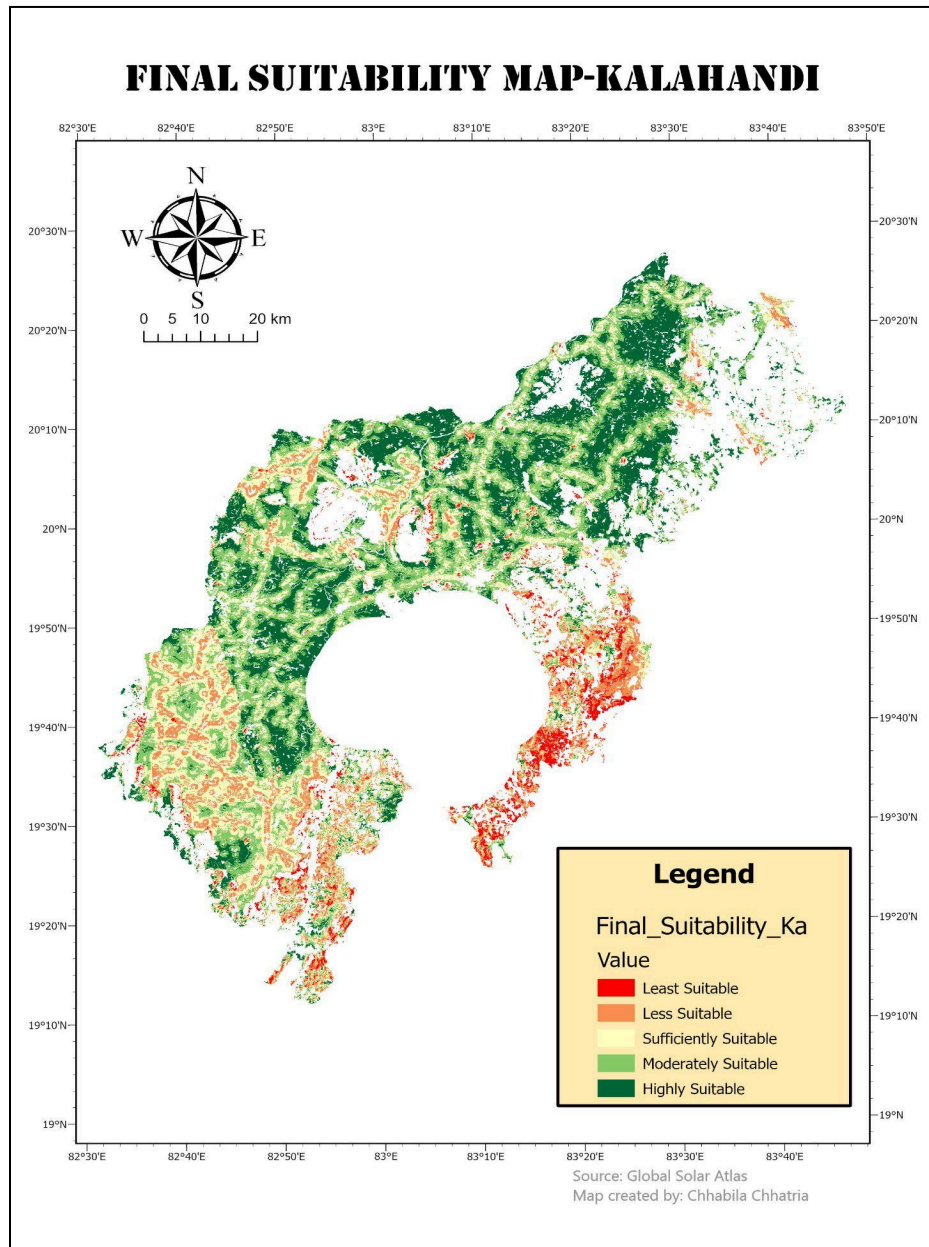
Before MCE analysis, all the factor rasters were reclassified on a 1-5 suitability scale, and restricted areas were completely excluded by applying a constraints mask. And all raster layers were aligned to the same projection and cell size to ensure pixel-wise computation.

By using the spatial Analyst weighted sum tool in ArcGIS Pro, each factor was multiplied by its AHP-derived weight, and all the weighted layers were added pixel by pixel to generate a composite Solar suitability index.

$$\text{Suitability Index} = (0.33 \times \text{GHI}) + (0.25 \times \text{Slope}) + (0.15 \times \text{Distance from road}) + (0.14 \times \text{Distance from built-up area}) + (0.08 \times \text{Elevation}) + (0.05 \times \text{Aspect})$$

The final Output of this analysis was a continuous raster, where each pixel's suitability score represented the combined effect of all criteria. The final output is again reclassified into High, Moderate, and Low suitability classes to make the results more interpretable for decision-making.

7.10) Final Suitability Map:



Shows that the high suitability zones are mostly concentrated in the central and western plains of the district. These areas have comparatively high solar radiation, and low slopes are located mainly close to roads and built-up areas. In the eastern and southeastern parts, where the terrain is more hilly and there is more forest cover, the suitability is comparatively low.

Results:

1. What is the total area available in the Kalahandi district that is suitable for solar projects?

In Kalahandi district, the total potential area available for solar power development that comes under the categories of high suitability and moderate suitability is 4448.87km².

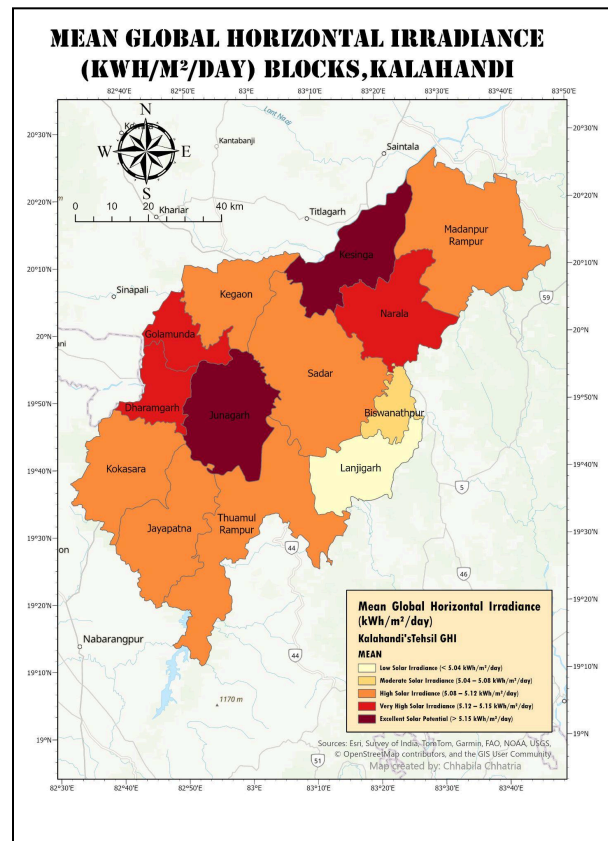
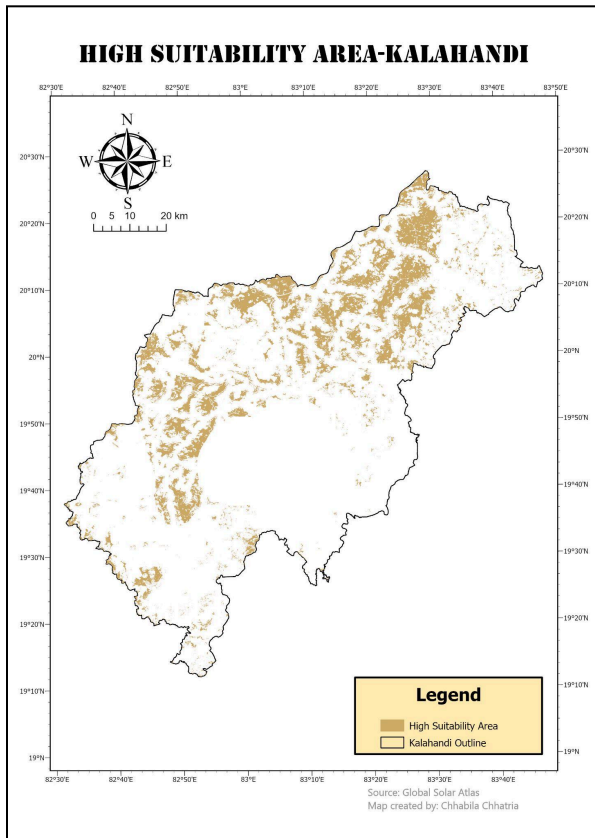
And if we consider only High suitability land, then the area is 1195.21km².

2. What percentage of the total land of the district is highly suitable?

$$\text{percentage} = 1195/7920 \times 100 \\ = 15.09\%$$

~15% of the area of Kalahandi is highly suitable.

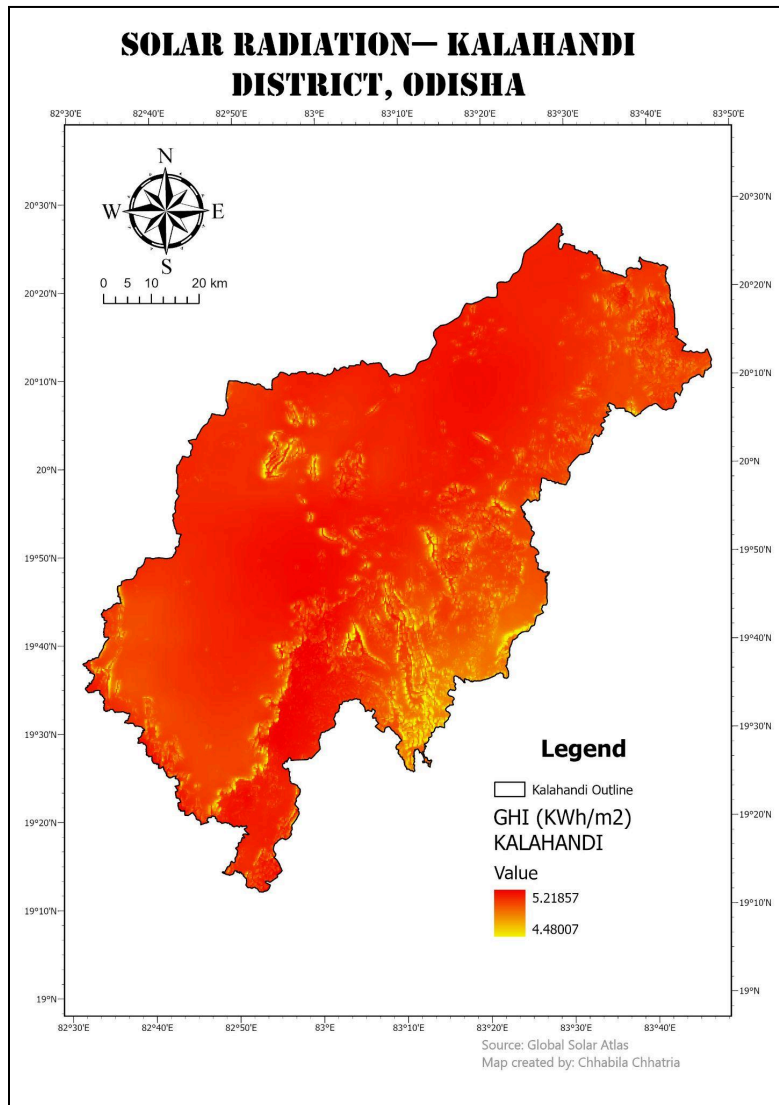
3. What are the variations in solar radiation received in different areas of the district, and how can we determine which type of PV panels to install to get the maximum productivity based on the particular features of each region (for example, regular, high concentration)?



Variation of GHI has also been found in the Kalahandi district, with tehsil-wise average values of GHI being from ~5.03 to 5.16kWh/m²/day.

Intensity of incoming solar radiation is mainly high in the tehsils of the district, like Junagarh, Kesinga, and Narala, whereas in areas dominated by hills, like Lanjigarh and Biswanathpur, slightly lower solar radiation is received.

Taking into account these differences, it can be proposed that conventional crystalline silicon solar panels are found to be most appropriate for Kalahandi, as they function effectively under high and moderately illuminated conditions. The application of high-concentration solar panels appears limited at a district level.



Discussion and conclusion:

In this paper, the suitability of solar energy development within the Kalahandi district was evaluated using a Multi-Criteria Evaluation with a GIS approach. A solar suitability map was developed using a Weighted Sum technique, where the criteria weights were obtained using an AHP. It is evident from this study that the solar potential of a region is not homogeneously distributed, but it is dependent on topography, land use, solar intensity, and accessibility.

≈ 15% of the land in Kalahandi is very suitable. Analysis revealed that about 4,448 km² could be identified as potentially suited for solar projects, of which 1,195 km² is of higher suitability, constituting about 15.09% of the total area. These high suitability areas mostly cover the central and western parts of the district, where higher GHI, gentle slope, and road connectivity reasonably exist. The areas that showed low suitability were mostly found in the eastern and southern parts of the study area, considering the existence of hilliness, forests, and protected areas.

Tehsil-wise GHI analysis has been able to verify that Kalahandi is a high solar potential area overall, with average GHI levels varying between 5.03-5.16 kWh/sq. meter/day. Taking into consideration the solar radiation levels and geographic conditions prevailing there, the research indicates that conventional Crystalline Silicon Photovoltaic Cells are the most feasible technology for Kalahandi, and High Concentrated Photovoltaic Cells are not appropriate at the tehsil level.

The results of this study are very much in sync with reality and also previous studies related to suitability analysis through GIS-based approaches, and offer a very scientific approach to planning at a district level. The suitability analysis may be improved in future studies by incorporating finer-level data.

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